

4.1 Quantitative Progress Metrics

The following measurable achievements were accomplished this week:

- Hardware Components: 100% of core drone components acquired and documented
- AI Model Performance: Achieved mAP@0.5 of 56.6% on rice disease validation set
- Dataset Expansion: Integrated 2,500+ annotated images for rice and wheat diseases
- System Integration: Established complete data pipeline (Flutter → Backend → AI → Firebase)
- Testing Coverage: Validated system with 50+ test images across different conditions
- Model Optimization: Reduced inference time to 50-80ms per image
- Multi-Crop Support: Successfully extended system from single-crop (rice) to dual-crop (rice + wheat)

These metrics demonstrate substantial progress toward a production-ready agricultural monitoring system.